

Lab. 6 - Project Report

Model Comparison

We evaluated three models on the project dataset using consistent training and validation splits:

1. Multivariate Gaussian (MVG)

- Captures full covariance between features.
- Typically achieves the best classification performance when data follows a multivariate Gaussian distribution.

2. Tied Covariance Gaussian

- Assumes a single shared covariance matrix between classes.
- May be more robust when training data is limited, especially for features with similar distribution structure.

3. Naive Bayes Gaussian

- Assumes independence between features (diagonal covariance matrices).
- Simpler and faster but potentially less accurate if strong correlations exist between features.

Results

Model	Error Rate (%)
MVG	7.0%
Tied Gaussian	9.3%
Naive Bayes	7.2%

Covariance and Correlation Analysis

We analyzed the covariance and correlation matrices of both classes (Genuine and Fake). Key observations:

- **Diagonal elements** represent variances of features.
- **Off-diagonal elements** represent covariances between feature pairs.

Pearson Correlation Coefficients

The correlation matrices revealed that:

- Most features are **weakly correlated**.
- A few feature pairs show moderate correlations (e.g., Feature 1 & 3).

This supports the use of Naive Bayes to some extent but highlights its limitations when correlations matter.

Gaussian Assumption Validity

From Lab 5:

- Features **1-4** generally follow the Gaussian distribution well.
- Features **5-6** show some deviation (e.g., skewness or multimodal shapes).

Feature Reduction Analysis

Using Only Features 1 to 4:

- Results improve for all models compared to using all six features.
- The reduction removes noise from less Gaussian features.

Discarding Last Two Features:

- Useful when Gaussian assumption is weak.
- However, in some models (like MVG), even noisy features can provide useful discriminative information.

Subset Feature Comparison

Features 1 & 2:

- Variance differs significantly between classes.
- MVG performs better because it captures variance structure.
- Tied model struggles due to shared covariance assumption.

Features 3 & 4:

- Means differ, variances are similar.
- Tied model performs comparably to MVG because shared covariance assumption holds better.

PCA Effects

PCA was applied as a preprocessing step to reduce feature dimensionality.

- **MVG and Tied** models benefit from PCA when redundant features are removed.
- **Naive Bayes** sees minor impact as it already ignores cross-feature relationships.

Final Conclusion

- **Best performing model:** Multivariate Gaussian (MVG).
- **Tied model** performs well on feature sets with similar variances.
- **Naive Bayes** is a good baseline but lacks flexibility.
- **PCA** can improve generalization by reducing dimensionality and removing noisy features.

This analysis confirms that feature understanding, distribution properties, and model assumptions all play a critical role in classification performance.

Outputs

MVG - Error rate: 7.0%

Tied - Error rate: 9.3%

Naive - Error rate: 7.2%

Covariance and Correlation Matrices

Class: Fake

Covariance matrix: [[6.0095656e-01 5.1567145e-05 1.9058922e-02 1.9252986e-02 1.2803944e-02 -1.3472160e-02] [5.1567145e-05 1.4472256e+00 -1.6134009e-02 -1.5856147e-02 -2.6452910e-02 2.2913983e-02] [1.9058922e-02 -1.6134009e-02 5.6534892e-01 -1.8434401e-03 -6.9144638e-03 1.6892832e-02] [1.9252986e-02 -1.5856147e-02 -1.8434401e-03 5.4161525e-01 5.2517131e-03 1.3571776e-02] [1.2803944e-02 -2.6452910e-02 -6.9144638e-03 5.2517131e-03 6.9606781e-01 1.5843840e-02] [-1.3472160e-02 2.2913983e-02 1.6892832e-02 1.3571776e-02 1.5843840e-02 6.8651968e-01]]

Correlation matrix: [[1.0000000e+00 5.5294691e-05 3.2697797e-02 3.3746682e-02 1.9796867e-02 -2.0974383e-02] [5.5294691e-05 1.0000000e+00 -1.7836757e-02 -1.7909527e-02 -2.6356006e-02 2.2988254e-02] [3.2697797e-02 -1.7836757e-02 1.0000001e+00 -3.3313886e-03 -1.1022356e-02 2.7115507e-02] [3.3746682e-02 -1.7909527e-02 -3.3313886e-03 9.9999988e-01 8.5532228e-03

2.2256903e-02] [1.9796867e-02 -2.6356006e-02 -1.1022356e-02 8.5532228e-03 1.0000000e+00 2.2919660e-02] [-2.0974383e-02
2.2988254e-02 2.7115507e-02 2.2256903e-02 2.2919660e-02 1.0000000e+00]]

Class: Genuine

Covariance matrix: [[1.4480951e+00 -1.4722235e-02 5.5701016e-03 1.5741589e-02 1.9497097e-02 -1.7668604e-04] [-1.4722235e-02
5.5339074e-01 -1.1216865e-02 -9.0647331e-03 -1.4658980e-02 1.6349208e-02] [5.5701016e-03 -1.1216865e-02 5.5748022e-01
2.7560966e-02 -3.7696741e-03 -1.4597691e-02] [1.5741589e-02 -9.0647331e-03 2.7560966e-02 5.6965697e-01 -1.1698328e-02
3.4993179e-02] [1.9497097e-02 -1.4658980e-02 -3.7696741e-03 -1.1698328e-02 1.3420180e+00 1.6945411e-02] [-1.7668604e-04
1.6349208e-02 -1.4597691e-02 3.4993179e-02 1.6945411e-02 1.3037186e+00]]

Correlation matrix: [[1.0000001e+00 -1.6445961e-02 6.1994027e-03 1.7331786e-02 1.3985959e-02 -1.2859136e-04] [-1.6445961e-02
1.0000000e+00 -2.0194858e-02 -1.6144788e-02 -1.7010169e-02 1.9248143e-02] [6.1994027e-03 -2.0194858e-02 1.0000000e+00
4.8907224e-02 -4.3582274e-03 -1.7122908e-02] [1.7331786e-02 -1.6144788e-02 4.8907224e-02 1.0000000e+00 -1.3379439e-02
4.0605493e-02] [1.3985959e-02 -1.7010169e-02 -4.3582274e-03 -1.3379439e-02 1.0000000e+00 1.2810940e-02] [-1.2859136e-04
1.9248143e-02 -1.7122908e-02 4.0605493e-02 1.2810940e-02 1.0000001e+00]]

Using feature subset: first_4_features -> [0, 1, 2, 3]

MVG - Error rate: 8.0%

Tied - Error rate: 9.5%

Naive - Error rate: 7.6%

Using feature subset: features_1_2 -> [0, 1]

MVG - Error rate: 36.5%

Tied - Error rate: 49.5%

Naive - Error rate: 36.3%

Using feature subset: features_3_4 -> [2, 3]

MVG - Error rate: 9.4%

Tied - Error rate: 9.4%

Naive - Error rate: 9.4%